



# Mobile Communications

## Chapter 7: Wireless LANs

- ❑ Characteristics
- ❑ IEEE 802.11
  - ❑ PHY
  - ❑ MAC
  - ❑ Roaming
- ❑ HIPERLAN
  - ❑ Standards
  - ❑ PHY
  - ❑ MAC
  - ❑ Ad-hoc networks
- ❑ Bluetooth





# Characteristics of wireless LANs

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## Advantages

- ❑ very flexible within the reception area
- ❑ Ad-hoc networks without previous planning possible
- ❑ (almost) no wiring difficulties (e.g. historic buildings, firewalls)
- ❑ more robust against disasters like, e.g., earthquakes, fire - or users pulling a plug...

## Disadvantages

- ❑ typically very low bandwidth compared to wired networks (1-10 Mbit/s)
- ❑ many proprietary solutions, especially for higher bit-rates, standards take their time (e.g. IEEE 802.11)
- ❑ products have to follow many national restrictions if working wireless, it takes a vary long time to establish global solutions like, e.g., IMT-2000





# Design goals for wireless LANs

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- ❑ global, seamless operation
- ❑ low power for battery use
- ❑ no special permissions or licenses needed to use the LAN
- ❑ robust transmission technology
- ❑ simplified spontaneous cooperation at meetings
- ❑ easy to use for everyone, simple management
- ❑ protection of investment in wired networks
- ❑ security (no one should be able to read my data), privacy (no one should be able to collect user profiles), safety (low radiation)
- ❑ transparency concerning applications and higher layer protocols, but also location awareness if necessary





# Comparison: infrared vs. radio transmission

## Infrared

- ❑ uses IR diodes, diffuse light, multiple reflections (walls, furniture etc.)

## Advantages

- ❑ simple, cheap, available in many mobile devices
- ❑ no licenses needed
- ❑ simple shielding possible

## Disadvantages

- ❑ interference by sunlight, heat sources etc.
- ❑ many things shield or absorb IR light
- ❑ low bandwidth

## Example

- ❑ IrDA (Infrared Data Association) interface available everywhere

## Radio

- ❑ typically using the license free ISM band at 2.4 GHz

## Advantages

- ❑ experience from wireless WAN and mobile phones can be used
- ❑ coverage of larger areas possible (radio can penetrate walls, furniture etc.)

## Disadvantages

- ❑ very limited license free frequency bands
- ❑ shielding more difficult, interference with other electrical devices

## Example

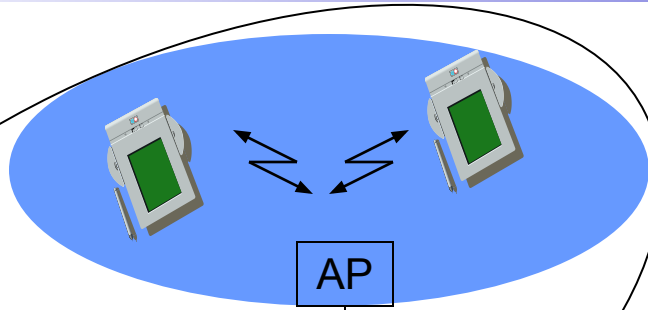
- ❑ WaveLAN, HIPERLAN, Bluetooth





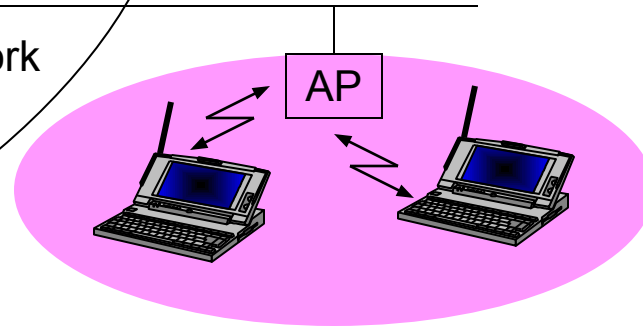
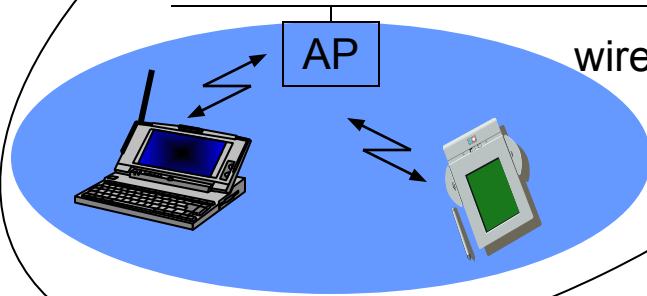
# Comparison: infrastructure vs. ad-hoc networks

infrastructure  
network

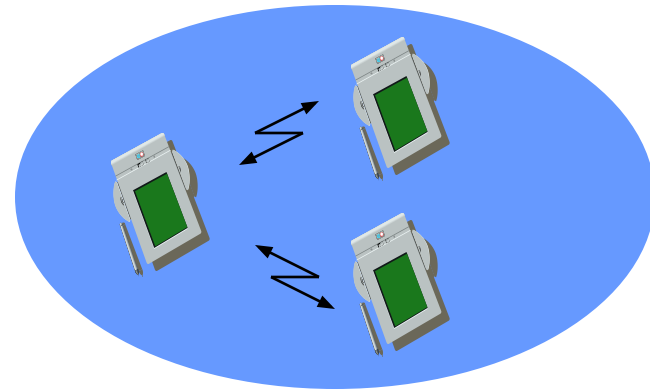
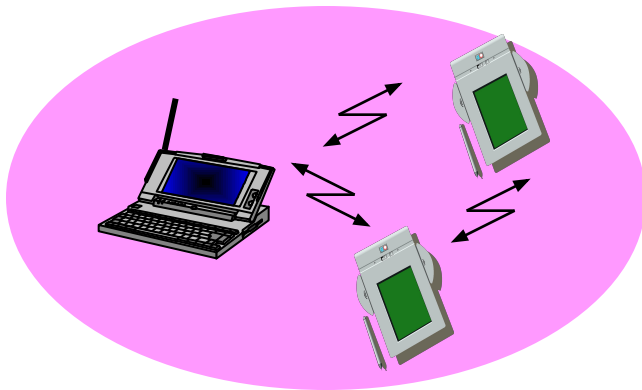


AP: Access Point

wired network

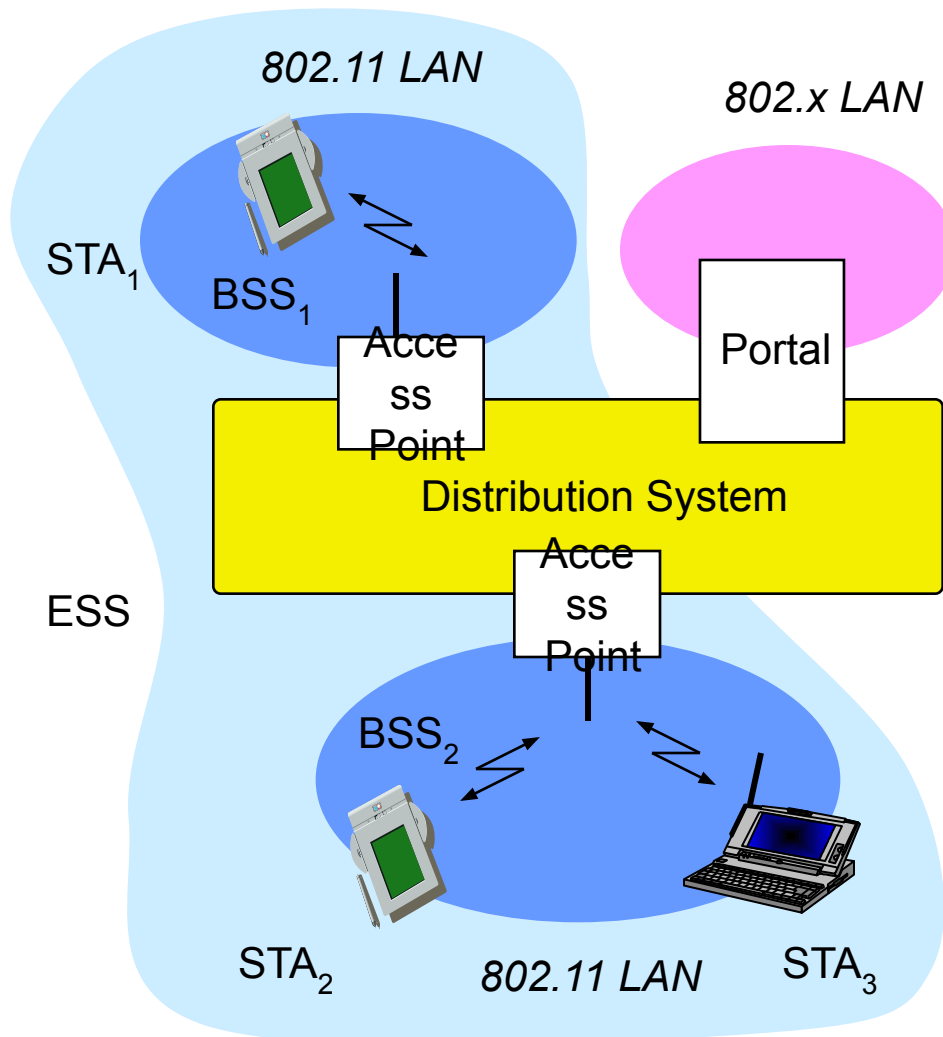


ad-hoc network





# 802.11 - Architecture of an infrastructure network



## Station (STA)

- terminal with access mechanisms to the wireless medium and radio contact to the access point

## Basic Service Set (BSS)

- group of stations using the same radio frequency

## Access Point

- station integrated into the wireless LAN and the distribution system

## Portal

- bridge to other (wired) networks

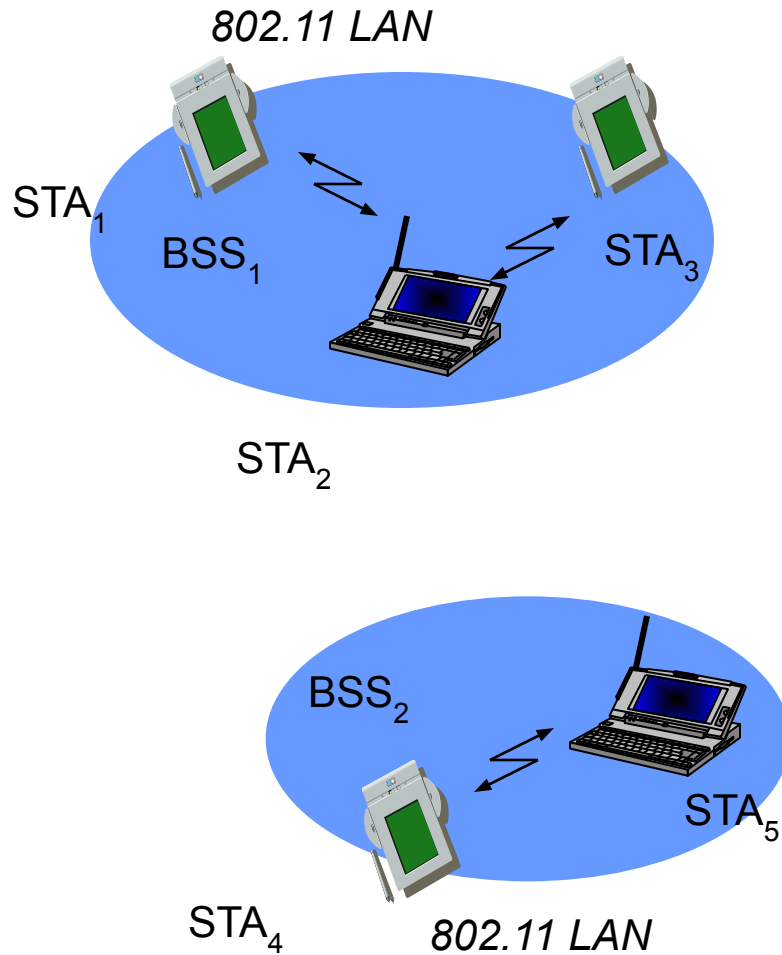
## Distribution System

- interconnection network to form one logical network (EES: Extended Service Set) based on several BSS





# 802.11 - Architecture of an ad-hoc network



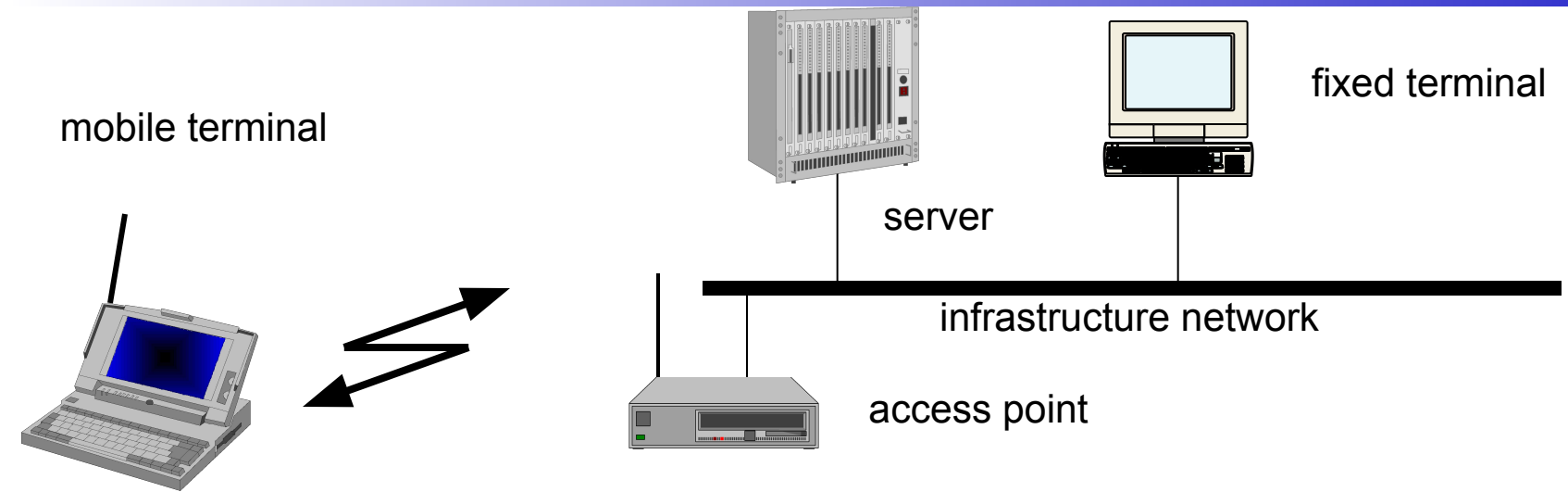
Direct communication within a limited range

- ❑ Station (STA): terminal with access mechanisms to the wireless medium
- ❑ Basic Service Set (BSS): group of stations using the same radio frequency





# IEEE standard 802.11



|             |
|-------------|
| application |
| TCP         |
| IP          |
| LLC         |
| 802.11 MAC  |
| 802.11 PHY  |

|            |           |
|------------|-----------|
| LLC        |           |
| 802.11 MAC | 802.3 MAC |
| 802.11 PHY | 802.3 PHY |

|             |
|-------------|
| application |
| TCP         |
| IP          |
| LLC         |
| 802.3 MAC   |
| 802.3 PHY   |







# 802.11 - Layers and functions

## MAC

- ❑ access mechanisms, fragmentation, encryption

## MAC Management

- ❑ synchronization, roaming, MIB, power management

## PLCP Physical Layer Convergence Protocol

- ❑ clear channel assessment signal (carrier sense)

## PMD Physical Medium Dependent

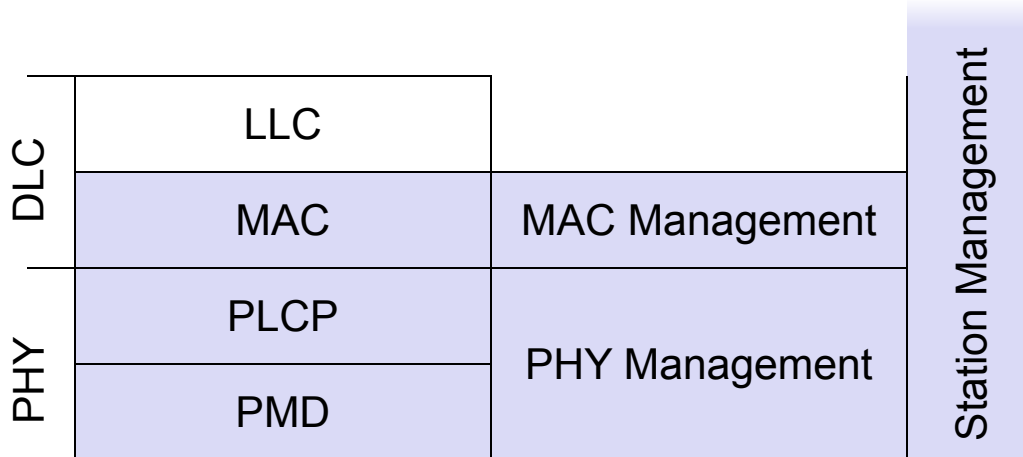
- ❑ modulation, coding

## PHY Management

- ❑ channel selection, MIB

## Station Management

- ❑ coordination of all management functions





# Future developments

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## IEEE 802.11a

- ❑ compatible MAC, but now 5 GHz band
- ❑ transmission rates up to 20 Mbit/s
- ❑ close cooperation with BRAN (ETSI Broadband Radio Access Network)

## IEEE 802.11b

- ❑ higher data rates at 2.4 GHz
- ❑ proprietary solutions already offer 10 Mbit/s

## IEEE WPAN (Wireless Personal Area Networks)

- ❑ market potential
- ❑ compatibility
- ❑ low cost/power, small form factor
- ❑ technical/economic feasibility
  - ❑ Bluetooth





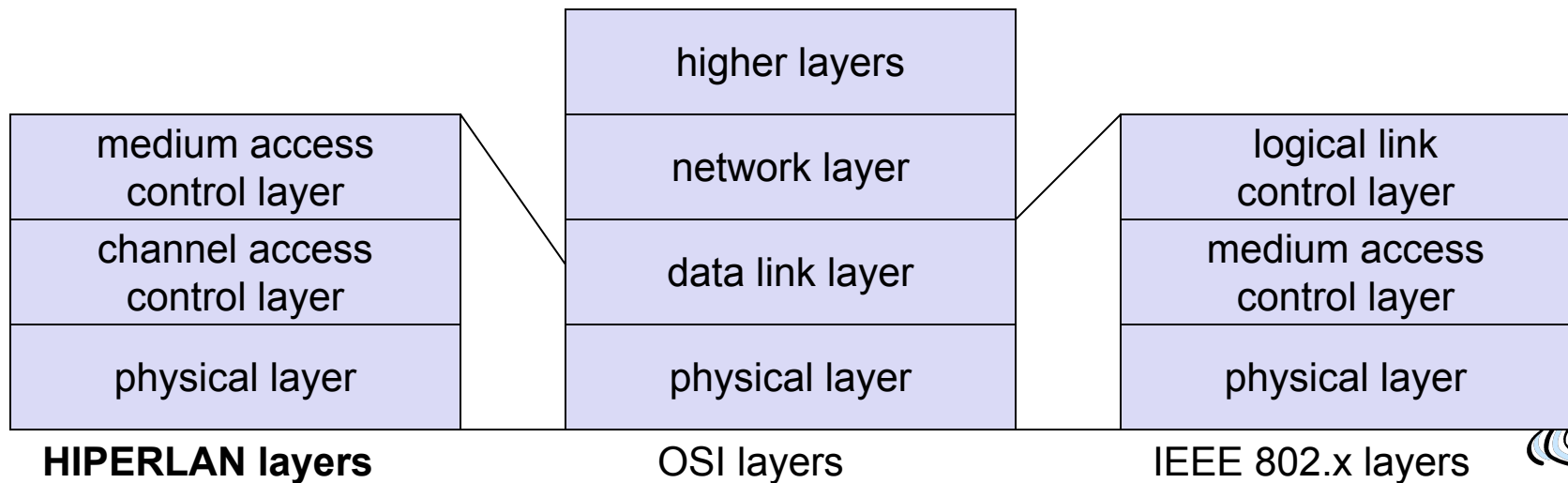
# ETSI - HIPERLAN

## ETSI standard

- ❑ European standard, cf. GSM, DECT, ...
- ❑ Enhancement of local Networks and interworking with fixed networks
- ❑ integration of time-sensitive services from the early beginning

## HIPERLAN family

- ❑ one standard cannot satisfy all requirements
  - range, bandwidth, QoS support
  - commercial constraints
- ❑ HIPERLAN 1 standardized since 1996





# Overview: original HIPERLAN protocol family

|                    | HIPERLAN 1                          | HIPERLAN 2                               | HIPERLAN 3          | HIPERLAN 4                              |
|--------------------|-------------------------------------|------------------------------------------|---------------------|-----------------------------------------|
| Application        | wireless LAN                        | access to ATM fixed networks             | wireless local loop | point-to-point wireless ATM connections |
| Frequency          | 5.1-5.3GHz                          |                                          |                     | 17.2-17.3GHz                            |
| Topology           | decentralized ad-hoc/infrastructure | cellular, centralized                    | point-to-multipoint | point-to-point                          |
| Antenna            | omni-directional                    |                                          | directional         |                                         |
| Range              | 50 m                                | 50-100 m                                 | 5000 m              | 150 m                                   |
| QoS                | statistical                         | ATM traffic classes (VBR, CBR, ABR, UBR) |                     |                                         |
| Mobility           | <10m/s                              |                                          | stationary          |                                         |
| Interface          | conventional LAN                    | ATM networks                             |                     |                                         |
| Data rate          | 23.5 Mbit/s                         | >20 Mbit/s                               |                     | 155 Mbit/s                              |
| Power conservation | yes                                 |                                          | not necessary       |                                         |

Check out Wireless ATM for new names!





# Bluetooth

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Consortium: Ericsson, Intel, IBM, Nokia, Toshiba - many members

At the same time the Bluetooth development started, a study group within IEEE 802.11 discussed **wireless personal area networks (WPAN)** under the following five criteria:

- **Market potential**
- **Compatibility**
- **Distinct identity**
- **Technical feasibility**
- **Economic feasibility**

Obviously, Bluetooth fulfills these criteria so the WPAN group cooperated with the Bluetooth consortium.

IEEE founded its own group for WPANs, IEEE 802.15, in March 1999.





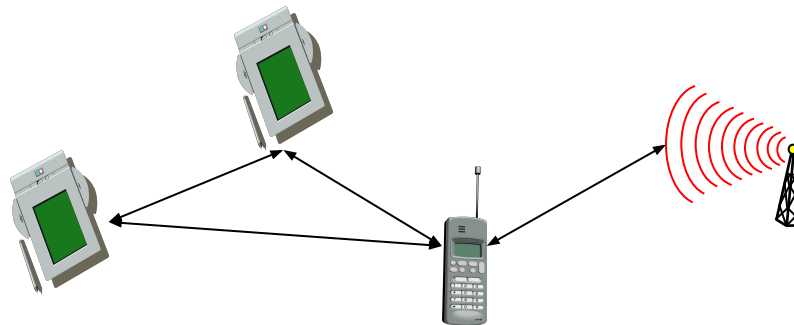
# Bluetooth

## Scenarios

- ❑ connection of peripheral devices
  - loudspeaker, joystick, headset
- ❑ support of ad-hoc networking
  - small devices, low-cost
- ❑ bridging of networks
  - e.g., GSM via mobile phone - Bluetooth - laptop

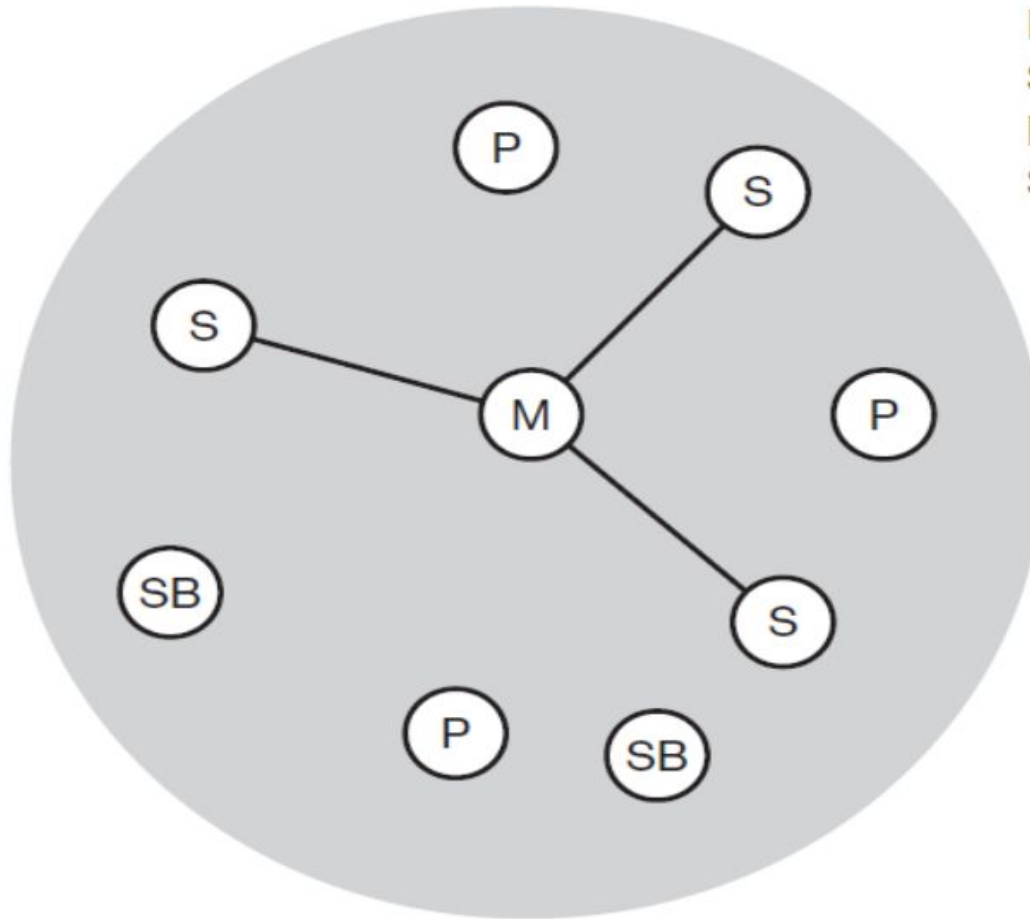
Simple, cheap, replacement of IrDA, low range, lower data rates

- ❑ 2.4 GHz, FHSS, TDD, CDMA





# Simple Bluetooth piconet

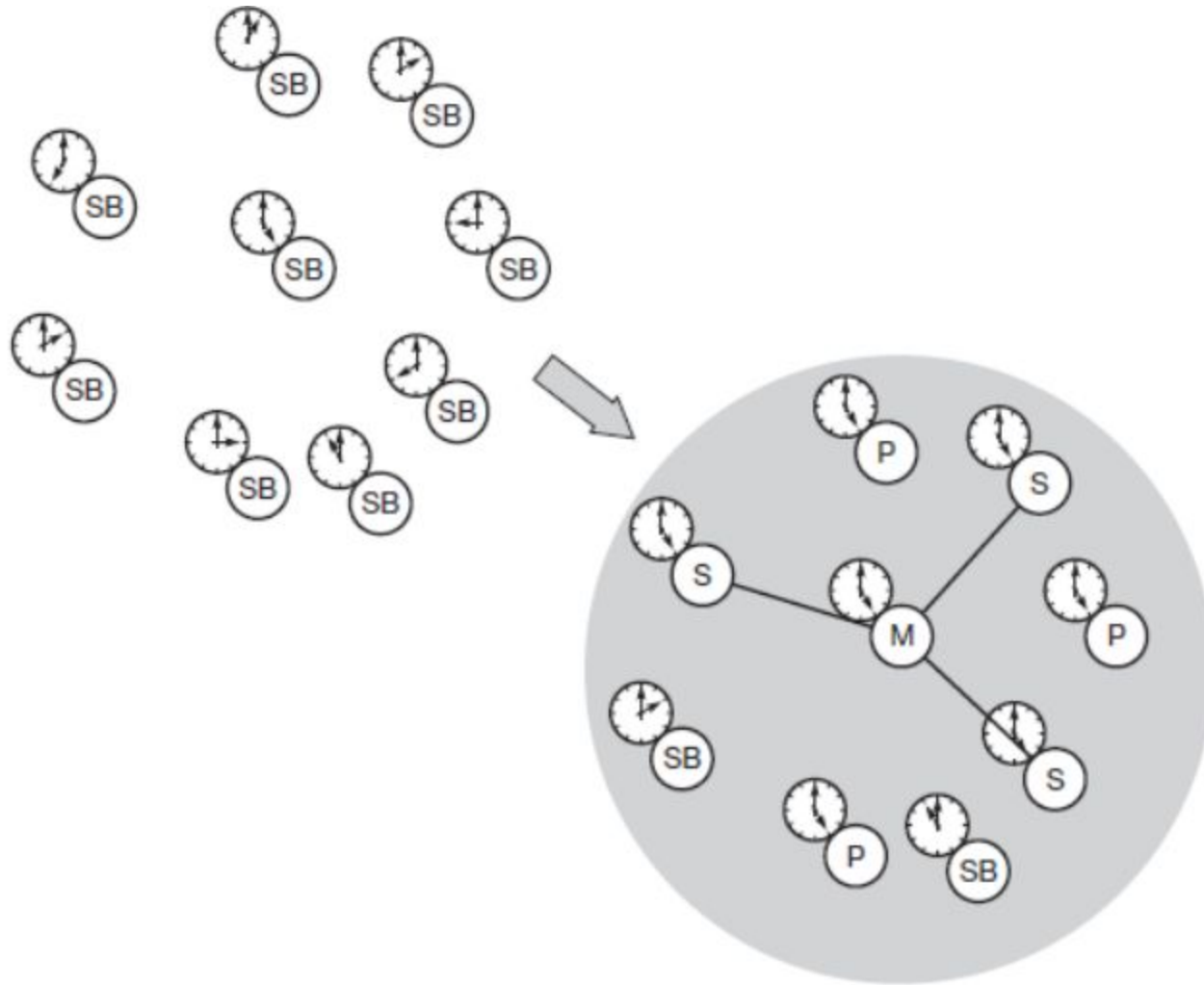


M = Master  
S = Slave  
P = Parked  
SB = Standby





# Forming a Bluetooth piconet







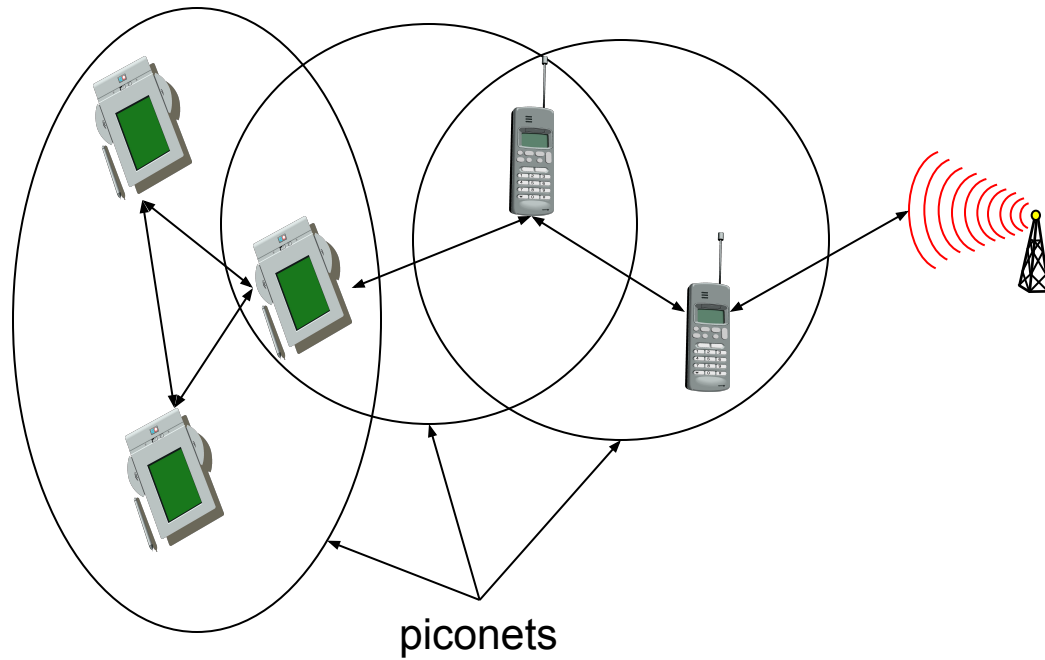
# Scatternets

Each piconet has one master and up to 7 slaves

Master determines hopping sequence, slaves have to synchronize

Participation in a piconet = synchronization to hopping sequence

Communication between piconets = devices jumping back and forth between the piconets





# Bluetooth scatternet

